

installed) must have a PXE network and public network.

Networks: The networks in Fuel are listed below.

Fuelweb_admin/PXE: This is used for PXE booting of all nodes in the environment, OS provisioning and hardware discovery.

Public network: This is used for the nodes' external communication.

Management network: This is for API and inter-service communication.

Storage network: This is for all storage communication like Swift, Cinder, etc.

Private network: This is the overlay network. All tenant virtual network communication happens over it. This network will be configured based on the network model you choose. For VLAN based segmentation, this network will not be created. This is required only for a tunnel network.

Storage requirements: Make sure the Fuel master has more than 60GB of disk space to install the OS and its services. In a production environment with hundreds of servers, it is recommended to have at least 512GB of storage as Fuel acts as a syslog server for all the nodes in the environment, which needs a lot of disk space. But with proper archiving and log rotation, this can be taken care of.

Installing Fuel

Download the latest Fuel ISO. As of writing this article, the latest available release is Fuel 9, which is based on OpenStack Mitaka. This can be downloaded from <https://www.mirantis.com/software/mirantis-openstack/releases/>

Configuring Fuel

Bring up the Fuel master with at least 60GB of disk space, two NICs, four CPUs and 8GB of RAM, with the first interface connected to the PXE network and the second interface connected to the public network (this interface will initially be used by all managed nodes to download packages from the Internet). Attach the ISO to the server, and follow the onscreen prompts to complete the installation, which will take less than 10 minutes. Fuel uses the CentOS 7 OS to run its services and has an automated installation procedure using Kickstart.

Once the installation is over, go to the console and configure *eth1* with the proper IP/subnet and gateway with



Figure 1: Installing Fuel



Figure 2: Configuring interface

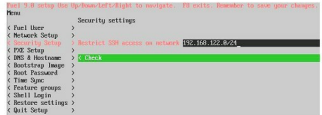


Figure 3: Configuring SSH access

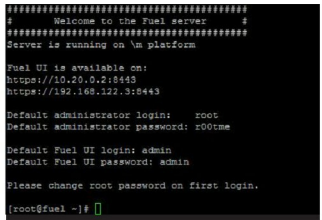


Figure 4: Console login

Internet access.

Configure the security set-up and add the subnet that is allowed to SSH to Fuel. If you set it to 0.0.0.0/0, all networks will have SSH access to Fuel, which is not recommended.

Once all the changes are done based on the environment, select *Quit setup*, and then *Save* and *Quit*.

It may take another 10 minutes to set up all the Fuel components and the bootstrap image. Once Fuel is completely configured based on the input provided, it will show a login prompt like what's shown in Figure 4.

The Bootstrap image is an Ubuntu 14.04 image with Nailgun and MCollective agents installed, with which Fuel will be able to detect new nodes and their hardware details.

Creating an environment in Fuel

Access <https://fuelip:8443> and log in as *admin/admin* (the default username and password, unless changed during deployment).

Click on 'New Openstack Environment' and then choose KVM/QEMU hypervisor (based on the type of hardware).

Select Neutron with the tunnelling segmentation and